

Sydney Technical High School



Mathematics Extension 1

H.S.C ASSESSMENT TASK 2

MARCH 2012

General Instructions

- Working Time – 70 minutes.
- Approved calculators may be used.
- A table of Standard Integrals is provided at the back of this paper.
- All necessary working should be shown for every question.
- Begin each question on a new side of the answer booklet.
- Marks shown are a guide and may need to be adjusted.
- Full marks may not be awarded for careless work or illegible writing.

NAME _____

TEACHER _____

Question 1

a) Find : i) $\int (7x - 2)^4 dx$

1

ii) $\int \frac{x+1}{\sqrt{x}} dx$

2

b) Use the substitution $u = 2 + x^2$, or otherwise, to evaluate $\int_0^1 \frac{x}{(2+x^2)^2} dx$

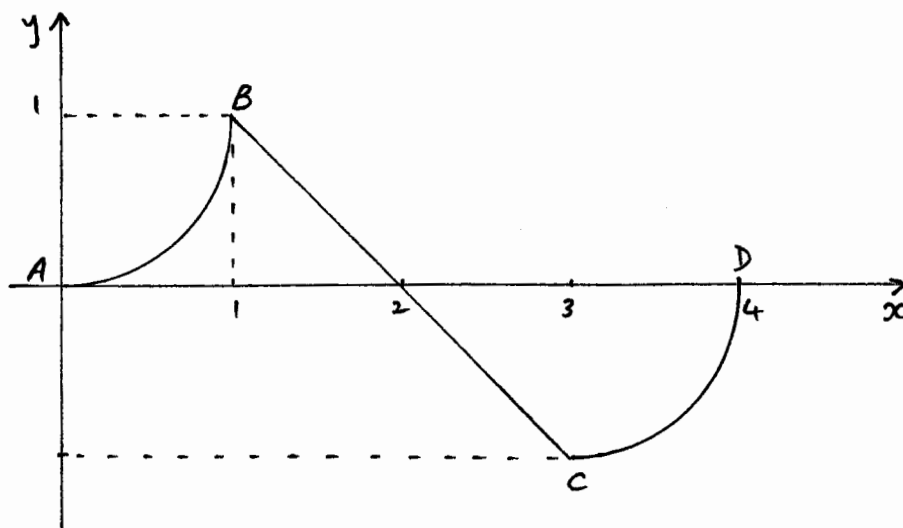
3

c) Solve $4 \cos^3 x - 3 \cos x = 0$ for $0 \leq x \leq 2\pi$

3

Question 2 (Start a new page)

- a) Below is shown the curve $y = f(x)$ for $0 \leq x \leq 4$. AB and CD are arcs of circles with centres $(0,1)$ and $(3,0)$ respectively.



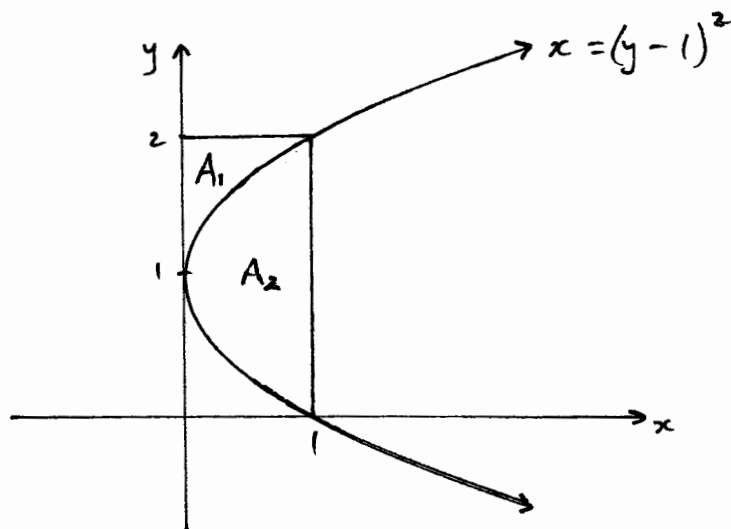
i) Evaluate $\int_0^1 f(x) dx$

1

ii) Evaluate $\int_0^4 f(x) dx$

1

b) Shown is the parabola $x = (y - 1)^2$



A_1 is the area bounded by the parabola, the y axis and $y = 2$.

A_2 is the area bounded by the parabola and $x = 1$.

- i) Find A_1 1
- ii) Find A_2 1
- iii) A_2 is rotated about the y -axis. Find the volume thus generated, in exact form. 3

c) Evaluate $\int_{-2}^2 \frac{x}{1+x^4} dx$ 1

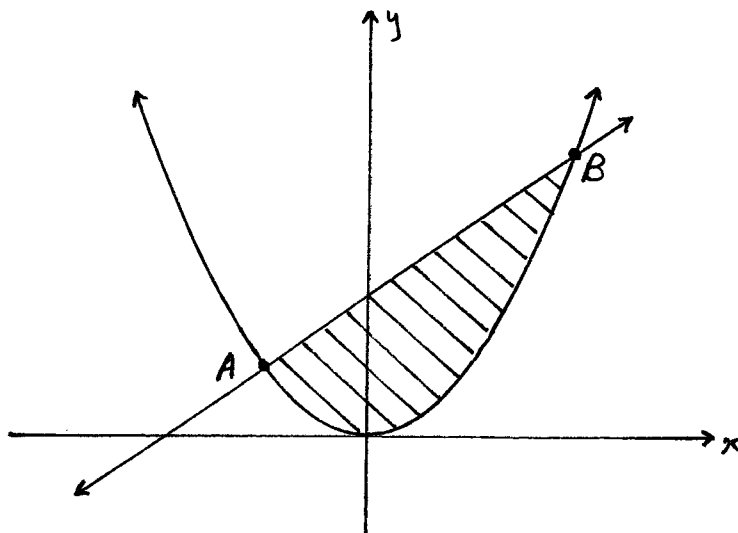
Question 3 (Start a new page)

- a) Find $\frac{d}{dx} [\sin(\tan x)]$ 1
- b) The data below gives values of $y = f(x)$ for $1 \leq x \leq 5$. 1

x	1	2	3	4	5
$f(x)$	0	1.4	3.3	2.8	1.5

Use the Trapezoidal Rule and 5 function values to approximate $\int_1^5 f(x) dx$.

c)



The area between the graphs of $y = x^2$ and $y = x + 2$ is shown. A and B are points of intersection of the two graphs.

i) Find x values for A and B .

1

ii) Find the value of the shaded area, correct to 1 dec. place

2

iii) The shaded area is rotated about the x -axis.

α) Write an integral expression to calculate the exact volume generated (do not evaluate).

1

β) Use Simpson's Rule and 3 function values to approximate the above volume. Leave your answer in simplest exact form.

3

Question 4 (Start a new page)

a) Simplify $\sin(\pi + \theta) \operatorname{cosec}(\pi - \theta)$

1

b) Given the curve represented by $y = 1 - x - \frac{1}{x-1}$.

i) Find y' and show that $y'' = \frac{-2}{(x-1)^3}$

1

ii) Locate and determine the nature of any stationary points.

3

iii) Locate any points of inflexion. Give reasons.

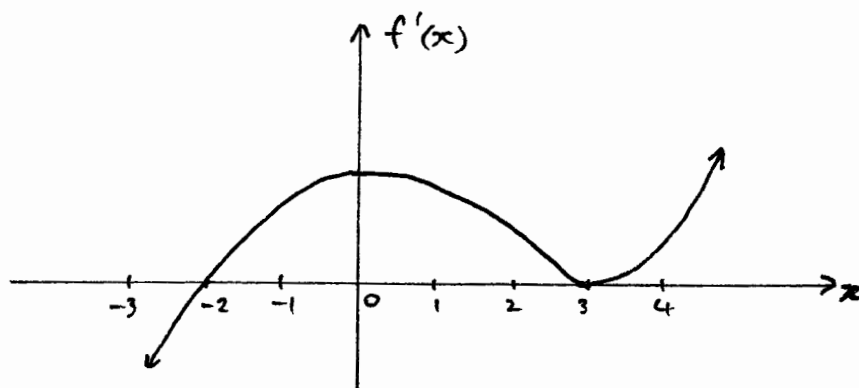
1

iv) Draw a neat sketch of the curve. Use a ruler for straight lines and label key features.

2

Question 5 (Start a new page)

a)



The diagram above shows the graph of $y = f'(x)$, i.e. the derivative curve of $y = f(x)$.

i) Give the locations and types of the stationary points on the curve $y = f(x)$.

2

ii) Which feature will appear on the curve $y = f(x)$ that corresponds to $x = 0$ above? Justify.

1

iii) For what values of x is the curve $y = f(x) : \alpha$ increasing ?

1

β) such that $f''(x) < 0$

1

iv) Neatly sketch a possible graph of $y = f(x)$, showing important x values.

1

v) Neatly sketch a possible graph of $y = f''(x)$, showing important x values.

1

b) i) Show that $\frac{d}{dx}(\sec^2 2x) = 4 \tan 2x \sec^2 2x$

2

ii) Hence, find $\int_0^{\frac{\pi}{3}} \tan 2x \sec^2 2x \, dx$

1

Question 6 (Start a new page)

a) i) On the same axes, neatly sketch and label the curves $y = \cos 3x$

2

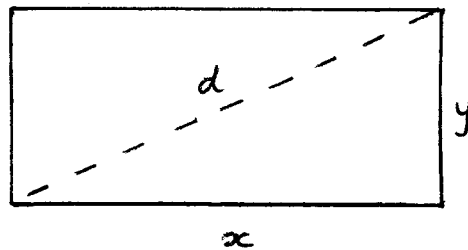
and $y = \cos 3\left(x - \frac{\pi}{6}\right)$ for $0 \leq x \leq \pi$. Clearly show intercepts on the axes.

ii) For what values of k will $\cos 3\left(x - \frac{\pi}{6}\right) = k$ have exactly 2 solutions, for $0 \leq x \leq \pi$?

1

b) A rectangle has fixed perimeter P cm. Its length, width and diagonal are variable and shown below.

4



Use calculus to prove that the shortest diagonal occurs when the rectangle is a square.

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Solutions.

1 a) i) $\frac{(7x-2)^5}{35} + c$

ii) $\int \left(\frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} \right) dx = \int (x^{\frac{1}{2}} + x^{-\frac{1}{2}}) dx$
 $= \frac{2}{3} x^{\frac{3}{2}} + 2x^{\frac{1}{2}} +$

b) $\int_0^1 \frac{x}{(2+x^2)^2} dx = \int_2^3 \frac{x}{u^2} \frac{du}{2x}$

$= \frac{1}{2} \int_2^3 u^{-2} du$

$= -\frac{1}{2} \left[\frac{1}{u} \right]_2^3$

$= -\frac{1}{2} \left(\frac{1}{3} - \frac{1}{2} \right)$

$= \frac{1}{12}$

$u = 2 + x^2$

$\frac{du}{dx} = 2x$

$dx = \frac{du}{2x}$

$x=0, u=2$

$x=1, u=3$

c) $\cos x (4\cos^2 x - 3) = 0$

$\cos x = 0$ or $\cos x = \pm \frac{\sqrt{3}}{2}$

$\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

2 a) i) $1 - \frac{\pi}{4}$ ii) $1 - \frac{\pi}{4} - \frac{\pi}{4} = 1 - \frac{\pi}{2}$

b) i) $A_1 = \int_1^2 (y-1)^2 dy$
 $= \left[\frac{(y-1)^3}{3} \right]_1^2$

$= \frac{1}{3} (1-0)$

$= \frac{1}{3} u^2$

ii) $A_2 = 2 - \frac{2}{3}$

$= \frac{4}{3} u^2$

iii) $Vol = \pi \times 1^2 \times 2 - 2\pi \int_0^1 (y-1)^4 dy$

$= 2\pi - 2\pi \left[\frac{(y-1)^5}{5} \right]_0^1$

$= 2\pi - 2\pi \left(0 - -\frac{1}{5} \right)$

$= 2\pi - \frac{2\pi}{5}$

$= \frac{8\pi}{5} u^3$

c) odd function $\Rightarrow$ answer is 0

③ a) $\cos(\tan x) \sec^2 x$

b) $\int_1^5 f(x) dx \doteq \frac{1}{2} (0 + 2 \cdot 8 + 6 \cdot 6 + 5 \cdot 6 + 1 \cdot 5)$
 $= 8.25$

c) i) $x^2 = x + 2$
 $x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0$

$\therefore x = 2 \text{ or } -1$

ii) $A = \left| \int_{-1}^2 (x^2 - x - 2) dx \right|$

$= \left| \left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_{-1}^2 \right|$

$= (\text{calc.})$

$= |-4 \frac{1}{2}|$

$= 4 \frac{1}{2} \text{ u}^2$

only if
necessary

iii) a) $\text{Vol} = \pi \int_{-1}^2 (x+2)^2 - x^4 dx$

$\beta) \text{Vol} \doteq \pi \times \frac{1 \frac{1}{2}}{3} \left[f(-1) + 4f(\frac{1}{2}) + f(2) \right]$
 $= \frac{\pi}{2} (0 + \frac{99}{4} + 0)$
 $= \frac{99\pi}{8} \text{ u}^3$

④ a) $-\cancel{\sin \theta} \times \frac{1}{\cancel{\sin \theta}} = -1$

b) i) $y = 1 - x - (x-1)^{-1}$
 $y' = -1 + (x-1)^{-2}$
 $y'' = -2(x-1)^{-3}$
 $= \frac{-2}{(x-1)^3}$

ii) S.P.'s when $y' = 0$

$$\therefore 1 = \frac{1}{(x-1)^2}$$

$$\therefore (x-1)^2 = 1$$

$$\therefore x-1 = \pm 1$$

$$\therefore x = 2 \text{ or } 0$$

①

$x=2, y'' < 0 \Rightarrow \text{max. T.P. at } (2, -2)$

$x=0, y'' > 0 \Rightarrow \text{min. T.P. at } (0, 2)$

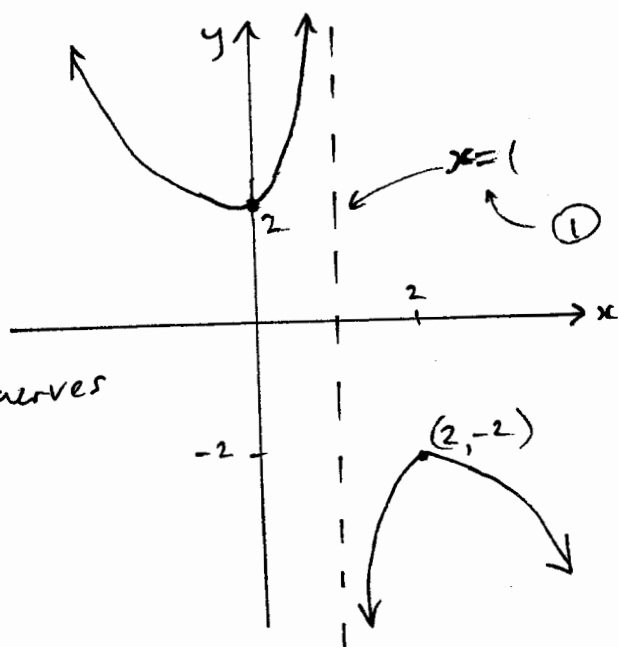
①

iii) P. of I if $y'' = 0$

but $\frac{-2}{(x-1)^3} = 0$ impossible

$\therefore$ no P. of inflexion.

iv)



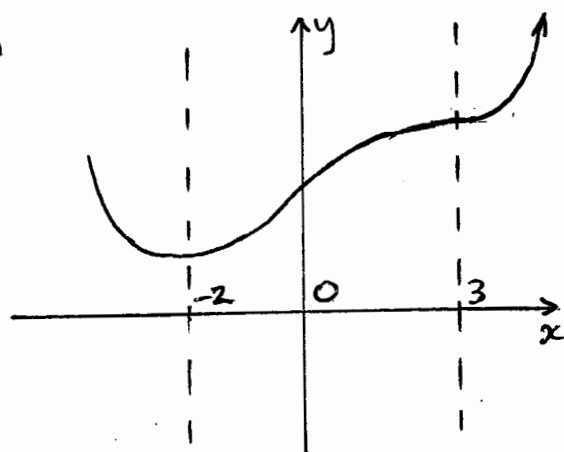
ii) pt. of inflexion, since there is max. pos. gradient when $x=0$ betw. the two stat. points. key point

iii) a) $-2 < x < 3$

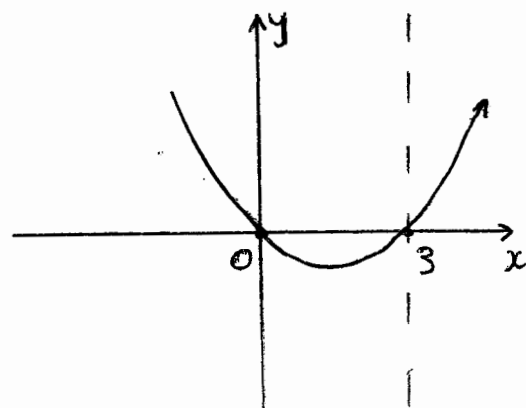
and $x > 3$ (both)

b) $f''(x) < 0$ means that $f'(x)$ is decreasing, ie. $0 < x < 3$.

iv)



v)



⑤ a) i) S.P. when $x = -2$ ~~not~~ ① minimum turn. pt.

S.P. when $x = 3$ ~~not~~ ① horizontal pt. of inflexion

$$b) i) \frac{d}{dx} (\sec^2 2x) = \frac{d}{dx} \left(\frac{1}{\cos^2 2x} \right)$$

$$= \frac{-2\cos 2x \times (-\sin 2x) \times 2}{(\cos^2 2x)^2} \quad \text{--- (1)}$$

$$= \frac{4\cos 2x \sin 2x}{\cos^4 2x}$$

$$= 4 \frac{\cancel{\cos 2x}}{\cancel{\cos 2x}} \times \frac{\cancel{\sin 2x}}{\cancel{\cos 2x}} \times \frac{1}{\cos^2 2x}$$

$$= 4 \tan 2x \sec^2 2x$$

need to show (1)

$$ii) = \frac{1}{4} [\sec^2 2x]_0^{\frac{\pi}{3}}$$

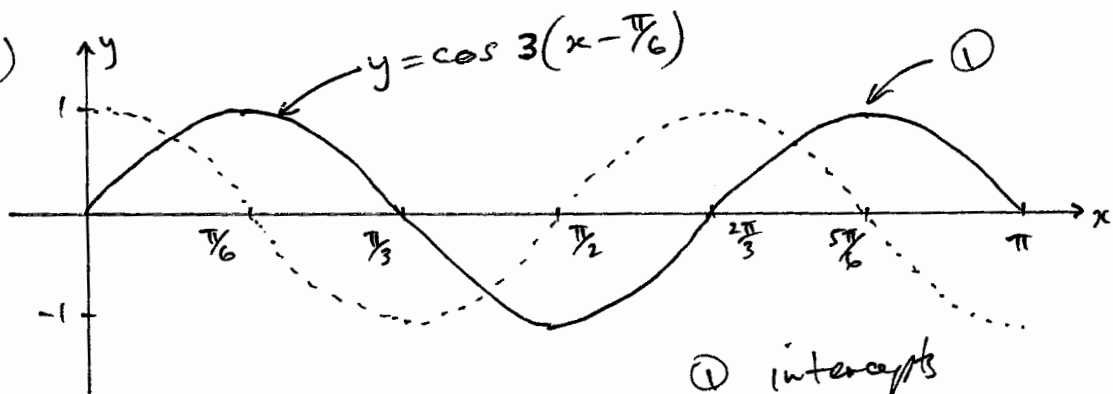
$$= \frac{1}{4} \left[\frac{1}{\cos^2 2x} \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{4} \left(\frac{1}{\frac{1}{4}} - \frac{1}{1} \right)$$

$$= \frac{1}{4} (4 - 1)$$

$$= \frac{3}{4}$$

(6) a) i)



$$ii) -1 < k < 0$$

$$b) d = \sqrt{x^2 + y^2}$$

$$= \sqrt{\left(\frac{p-2y}{2}\right)^2 + y^2}$$

$$= \sqrt{\frac{p^2 - 4py + 4y^2}{4} + \frac{4y^2}{4}}$$

$$= \frac{1}{2} \sqrt{p^2 - 4py + 8y^2} \leftarrow \textcircled{1} \text{ or similar}$$

minimum d when $d' = 0$

$$d' = \frac{1}{2} \times \frac{1}{2} (p^2 - 4py + 8y^2)^{-\frac{1}{2}} \times (16y - 4p) = 0 \leftarrow \textcircled{1}$$

$$\therefore \frac{16y - 4p}{4\sqrt{\quad}} = 0$$

$$\therefore 16y = 4p$$

$$\therefore y = \frac{p}{4} \leftarrow \textcircled{1}$$

necessary

y	$\frac{p}{5}$	$\frac{p}{4}$	$\frac{p}{3}$
d'	-	0	+



$\therefore$ minimum diagonal is proved

when $y = \frac{p}{4}$ and $x = \frac{p - \frac{2p}{4}}{2}$

$$= \frac{\frac{p}{2}}{2}$$

$$= \frac{p}{4} \quad \textcircled{1}$$

$\therefore$ equal sides in rectangle

$\therefore$ square is a rectangle.